

Course: B. Tech. Branch : Mechanical Engineering Semester : V

Subject Code & Name: BTMC 502 Machine Design - I

Max Marks: 60 Date: 31/01/23 Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following. 12

A) Write answers of following multiple choice questions. 1/2 6

- 1) Components made of brittle materials fails because of.....
(A) Sudden fracture (B) Yielding (C) Elastic deflection (D) All of the above
- 2) The following item(s) is (are) produced by the craft evolution process..
(A) Rowing boat (B) Musical instruments
(C) Bullock cart (D) All of the above
- 3) Ergonomics means....
(A) External appearance (B) Natural laws of work
(C) Streamline shape (D) All of the above
- 4) A sunk key fits in the keyway of the
(A) Hub only (B) Sleeve only
(C) Both hub and sleeve (D) Neither hub nor sleeve.
- 5) Woodruff key permits movement between shaft and the hub.
(A) Axial (B) Radial (C) Eccentric (D) None of the listed
- 6) A tension member of 40 mm diameter is to be replaced by a square of same material; side of square will be.....
(A) 40 mm (B) 36 mm (C) 60 mm (D) 25 mm

B) 1) Distinguish between design synthesis and design analysis. 1/2 6

2) What is standardization? Explain its importance in design.

C) Write short note on :- 1/2 6

- 1) Aesthetic considerations in design.
- 2) Factor of safety and its selection

Q.2 Solve Any Two of the following. 12

A) Explain the construction and working principal of cotter joint with the help of neat diagram. 2/2 6

B) The stress induced at a critical point in a machine component made of 45C8 ($S_{yt} = 380 \text{ N/mm}^2$) are as follows, 2/3 6

$$\sigma_x = 100 \text{ N/mm}^2 \quad \sigma_y = 40 \text{ N/mm}^2 \quad \tau = 80 \text{ N/mm}^2$$

Calculate factor of safety by

- 1) Maximum normal stress theory
- 2) Maximum shear stress theory
- 3) The distortion energy theory

- C) A wall bracket with a rectangular cross section is as shown in the Figure 1, the depth of the cross section is twice that of the width. The force P is acting on the bracket at 60 degree to the vertical is 6 kN. The material is made of Gray cast iron FG 250 and the factor of safety is 3. Determine the dimensions of the cross section of the bracket. Assume maximum normal stress theory of failure.

2/4

6

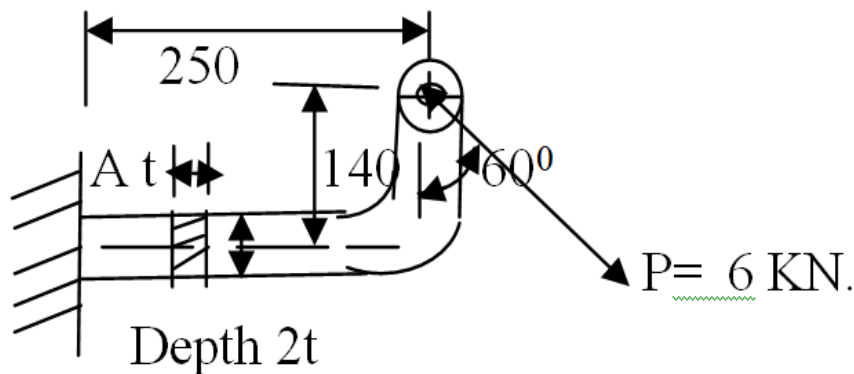


Figure 1. (All dimensions are in mm)

Q. 3 Solve Any Two of the following.

12

- A) 1) What is the stress concentration? Explain the various methods of reducing stress concentration.
2) What is fluctuating stress? Draw a stress – time curve for fluctuating stress.
3) Explain the terms fatigue failure & fatigue life?
- B) A plate made of steel 20C8 ($S_{ut} = 400 \text{ N/mm}^2$) in hot rolled and normalized condition is as shown in the Figure 2. It is subjected to completely reversed axial load of 30 kN. The notch sensitivity factor can be taken as 0.8 and the expected reliability is 0.85. The factor of safety is 2. Determine the plate thickness for infinite life. Use the below data if required
 $K_a = 0.67$, $K_b = 0.85$, $K_c = 0.897$, $K_t = 2.51$

1/2

6

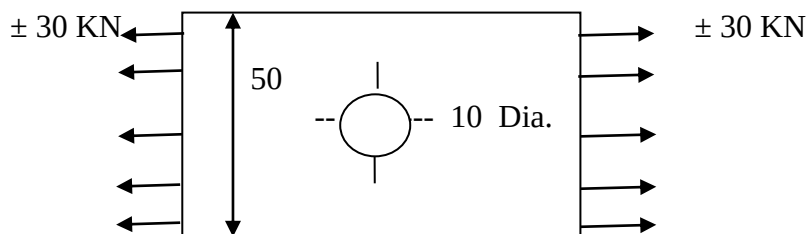


Figure 2. (All dimensions are in mm)

- C) A transmission shaft of cold drawn steel 27Mn2 ($S_{ut} = 500 \text{ N/mm}^2$ and $S_{yt} = 300 \text{ N/mm}^2$) is subjected to a fluctuating torque which varies from -100 N-m to + 400 N-m. If Factor of safety is 2 and expected reliability is 90 %. Neglecting the effect of stress concentration, determine the diameter of shaft assuming distortion theory of failure. Use the below data if required $K_a = 0.79$, $K_b = 0.85$, $K_c = 0.897$

2/5

6

Q.4 Solve Any Two of the following.

		12
A)	What is the function of keys? Explain the different types of keys along with their neat sketches.	1/2 6
B)	A shaft made of plain carbon steel 40C8 ($S_{yt} = 380 \text{ N/mm}^2$) is mounted with pulley 1000 mm in diameter overhang the left hand bearing by 250 mm. The belts mounted on it transmit the power to machine shaft below the pulley. Tension on tight side and slack side are 3 KN and 1 KN respectively. The weight of pulley is 500 N. The angle of wrap is 180 degree. If factor of safety is 3, determine the diameter of the shaft.	3/6 6
C)	A flexible coupling is used to transmit 15 KW at 100 rpm. There are 6 pins at pitch circle diameter (PCD) 200 mm. the effective length of the bush is 35 mm. the gap is 5 mm and pin length is 23 mm. The permissible shear and bending stress for the pin are 35 N/mm^2 & 152 N/mm^2 respectively. Calculate the pin diameter under shear and bending consideration.	2/6 6

Q. 5 Solve Any Two of the following.

		12
A)	1) What is the cause of residual stresses in welded joint? How are they relieved? 2) Explain the surge phenomena in spring and methods to avoid the same.	1/2 6
B)	A helical compression spring made of circular wire is subjected to an axial force which varies from 2.5 KN to 3.5 KN. Over this range of force, the deflection of the spring is 5 mm .the spring index can be taken as 5. The spring has square and ground ends. The spring is made of cold drawn steel wire with ultimate tensile strength of 1050 N/mm^2 and modulus of rigidity of 81370 N/mm^2 . The permissible shear strength for the spring wire should be taken as 50 % of the ultimate tensile strength. Design the spring and calculate a) Wire diameter, b) Mean coil diameter, c) Number of active coils, d) Total number of coils. e) Solid length of the spring, f) Free length of the spring.	3/6 6
C)	A single start square threaded screw is used in a screw press to exert a force of 75 KN. The screw is made of plain carbon steel 10C4 having ultimate strength = 350 N/mm^2 and factor of safety of 4. The permissible compressive stress is equal to permissible tensile stress and permissible shear stress is 50% of permissible tensile stress. The nut is made of grey cast iron FG200 and permissible bearing pressure between contacting surfaces of screw and nut is 17 MPa. A low friction thrust ball bearing is used in the mechanism and collar friction can be neglected. The coefficient of friction at the thread surface between steel screw and cast iron nut can be taken as 0.15. Determine the size	3/6 6

of screw and length of nut. The axial length of threads in the nut should be between 1- 1.5 times the nominal diameter of the screw. (Use the below data)

Sr. No	Nominal diameter, d (mm)	Pitch, p (mm)
1	30,32, 36	6
2	40 , 44	7

***** End *****